THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** a focused question	Questions are the starting point
2	**(a)** temperature of the oil	A relevant controllable factor
3	**(d)** making sense of the results later	Records are evidence
4	**(b)** Does the thickness of dough change how the puri puffs?	It can be tested by an experiment
5	**(c)** wonder through why and how questions	Science begins with wonder
6	**(d)** curiosity taking flight	Curiosity soars
7	**(c)** environment	Air, water, sunlight, organisms
8	**(a)** light	Chapter on light
9	**(b)** unit	Numbers need units
10	**(d)** new variable	A fresh question to test

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A factor that can change in an experiment (1). Oil temperature or dough thickness (1).	1+1
12	To know which change caused the effect (1). Other factors kept same keep the test fair (1).	1+1
13	Thickness of dough (1). Temperature of oil, type of flour (any one) (1).	1+1
14	Yes/no: does it puff? (1). Numerical: time to puff in seconds (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) Oil temperature (1). (ii) Time to puff or whether it puffs (1). (iii) Thickness, size, flour, way of dropping (1).	1+1+1
16	Controlled factors are set by us, e.g. dough thickness (1½). Observed factors are results, e.g. time to puff (1½).	1½+1½
17	Dough thickness (1). Type of flour (1). Oil temperature or way of dropping dough (1).	1+1+1
	OR: We cannot tell which change caused the result (1½). A fair test changes one variable at a time (1½).	1½+1½
18	Ask a focused question (1). Plan changing one factor and observing (1). Record, explain and ask new questions (1).	1+1+1
	OR: Moon phases repeat in a cycle (1). Observing sunrise, sunset and Moon cycles gave calendars (1). So earthly routines link to the sky (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) How does oil temperature affect puffing? (1). (ii) Oil temperature at three levels (1). (iii) Same dough, thickness, size, flour, dropping method (1). (iv) Whether it puffs and time to puff (1). (v) Table of trials with notes on splatter, smell and smoke (1).	1+1+1+1+1
	OR: A fair test changes one factor only (1). Example with puris (1). Controls keep other factors same (1). Otherwise cause cannot be known (1). Results become unreliable (1).	1+1+1+1+1
20	Notice a phenomenon (1). Ask a focused question (1). Plan experiments changing one thing (1). Observe, measure and record (1). Explain results and ask new questions (1).	1+1+1+1+1
	OR: Moon's lit part changes in phases (1). Phases repeat in a cycle (1). People counted days between cycles (1). Combined with sunrise and sunset observations calendars formed (1). Routines on Earth depend on sky motions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** fair (controlled) test	Only one variable is changed
2	**(c)** whether the puri puffs up	Only two answers are possible
3	**(b)** asks new questions	Each answer opens new questions
4	**(b)** Which colour is the nicest?	Opinions cannot be measured
5	**(b)** keeps evolving with new ideas	Ideas change as we learn
6	**(b)** cycles of the Sun and Moon	Phases and seasons
7	**(a)** why Earth is just right for life	Our home Earth
8	**(c)** observation, measurement and experiment	Scientific method
9	**(d)** stopwatch	Measures seconds
10	**(d)** starting an investigation	Curiosity-driven question

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They record every observation (1). They help in explaining results and planning new tests (1).	1+1
12	Asking a focused question (1). Changing one factor, observing and recording step by step (1).	1+1
13	Stopwatch measures time (1). Thermometer measures oil temperature (1).	1+1
14	Thickness also affects puffing (1). Keeping it same keeps the test fair (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) Oil temperature (1). (ii) Dough thickness, size, flour, dropping method (1). (iii) Time to puff (1).	1+1+1
16	Each answer reveals new unknowns (1½). Science keeps evolving as ideas are tested (1½).	1½+1½
17	Curiosity (1). Careful observation (1). Willingness to test and change ideas using evidence (1).	1+1+1
	OR: Many factors act together (1). Scientists still investigate it (1). It shows science is never finished (1).	1+1+1
18	Light, calendars/Moon, ecosystems (any three) (1½). Link: Moon phases led to calendars (1½).	1½+1½
	OR: Everyday events like dough rising can be investigated (1½). No fancy laboratory is needed to begin (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Dough thickness (1). (ii) Oil temperature, flour type (1). (iii) Puffs or not, time to puff (1). (iv) Adult handles hot oil, keep distance (1). (v) Does stored dough puff differently? (1).	1+1+1+1+1
	OR: Changed: amount of water (1½). Measured: height of plant after days (1½). Fixed: type of soil, light, pot size (2).	1½+1½+2
20	Light: mirrors and lenses explain images (1). Moon and Sun positions explain phases and calendars (1). Ecosystems show relationships of living things (1). Earth is just right for life (1). All use observation and measurement (1).	1+1+1+1+1
	OR: Records preserve evidence (1). They allow comparison of trials (1). Include dough size and thickness (1). Include oil temperature and time (1). Include sensory notes such as splatter or smoke (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.